Question Paper

## Physics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The electric field in a region is given by $\vec{E} = \frac{E_0 x}{l}\,\hat{i}$. The charge contained inside a cubical volume bounded by the surfaces $x = 0, x = l, y = 0, y = l, z = 0$ and $z = l$ is -

A. $2\varepsilon_0 E_0 l^2$

B. $\varepsilon_0 E_0 l^2$

C. $6\varepsilon_0 E_0 l^2$

D. zero

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The molar specific heat of a gas as given from the kinetic theory is $\frac{5}{2}R$. If it is not specified whether it is $C_p$ or $C_v$ , one could conclude that the molecules of the gas

A. Are definitely monoatomic

B. Are definitely rigid diatomic

C. Are definitely non-rigid diatomic

D. Can be monoatomic or rigid diatomic

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Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A particle of mass $M$ is moving in a horizontal circle of radius $R$ with uniform speed $V$. When the particle moves from one point to a diametrically opposite point, its

A. momentum does not change

B. momentum changes by $2MV$

C. kinetic energy changes by $\frac{MV^2}{4}$

D. kinetic energy changes by $MV^2$

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Diameter of a steel ball is measured using a Vernier callipers which has divisions of $0.1$ cm on its main scale (MS) and 10 divisions of its Vernier scale (VS) match 9 divisions on the main scale. Three such measurement for a ball are given as.

| S. No. | MS (cm) | VS divisions |
|---|---|---|
| 1. | 0.5 | 8 |
| 2. | 0.5 | 4 |
| 3. | 0.5 | 6 |

If the zero error is $-0.03$ cm, then mean corrected diameter is.

A. $0.53$ cm

B. $0.56$ cm

C. $0.59$ cm

D. $0.52$ cm

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A solid iron ball and an aluminium ball of same diameter are released together in a lake. Which ball will reach the bottom first. (neglect viscosity and $\rho_{Al} < \rho_{Fe}$)

A. Aluminium Ball

B. Iron Ball

C. Both will reach at the same time

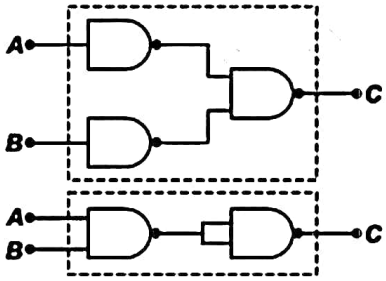
D. Data insufficient

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The combination of NAND gates shown here under figure, are equivalent to




A. an OR gate and an AND gate respectively

B. an AND gate and a NOT gate respectively

C. an AND gate and an OR gate respectively

D. an OR gate and a NOT gate respectively

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Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A double star system consists of two stars $A$ and $B$ which have time periods $T_A$ and $T_B$. Radius $R_A$ and $\mathrm{R_B}$ and mass $\mathrm{M_A}$ and $\mathrm{M_B}$. Choose the correct option.

A. If $T_A > T_B$ then $R_A > R_B$

B. If $T_A > T_B$ then $M_A > M_B$

C. $\left(\frac{T_A}{T_B}\right)^2 = \left(\frac{R_A}{R_B}\right)^3$

D. $\mathrm{T_A} = \mathrm{T_B}$

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The de Broglie wavelength of an electron moving with a velocity $1.5 \times 10^8\ \mathrm{ms}^{-1}$ is equal to that of a photon. The ratio of the kinetic energy of the electron to that of the energy of photon is :

A. 1.5

B. 1

C. 0.5

D. 0.25

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Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A student is at a distance 16 m from a bus when the bus begins to move with a constant acceleration of 9 m $s^{-2}$. The minimum velocity with which the student should run towards the bus so as the catch it is $\alpha\sqrt{2}$ $ms^{-1}$. The value of $\alpha$ is

A. 10

B. 12

C. 15

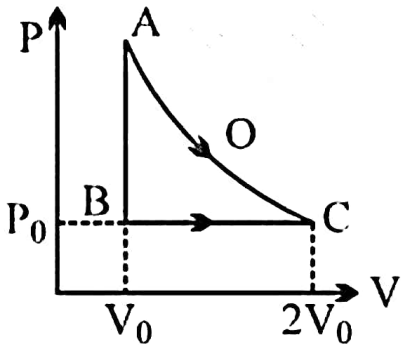
D. 20

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

An ideal gas is taken from point A to point C on P-V diagram through two process AOC and ABC as shown in the figure. Process AOC is isothermal




A. Process AOC requires more heat than process ABC.

B. Process ABC requires more heat than process AOC.

C. Both process AOC \& ABC require same amount of heat.

D. Data is insufficient for comparison of heat requirement for the two processes.

**Question No. 11**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

A battery of internal resistance $2\Omega$ is connected to a variable resistor whose value can vary from $4\Omega$ to $10\ \Omega$. The resistance is initially set at $4\Omega$. If the resistance is now increased then :

A. power consumed by it will decrease

B. power consumed by it will increase

C. power consumed by it may increase or may decrease

D. power consumed will first increase then decrease.

**Question No. 12**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

To estimate $g$ from $g = 4\pi^2 \frac{L}{T^2}$, error in measurement of $L$ is $\pm 2\%$ and error in measurement of $T$ is $\pm 3\%$. The error in estimated $g$ will be

A. $\pm 8\%$

B. $\pm 5\%$

C. $\pm 3\%$

D. $\pm 6\%$

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Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A machine which is $70\%$ efficient raises a $10$ kg body through a certain distance and spends $100$ J energy. The body is then released. On reaching the ground, the kinetic energy of the body will be

A. $0$

B. $70$ J

C. $50$ J

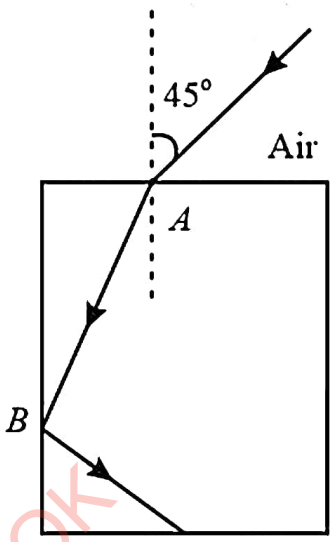
D. $35$ J

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A ray of light is incident at an angle of $45^\circ$ from air to a glass slab at point $A$, the angle of incidence being $45^\circ$.The minimum value of the refractive index of the material of the slab so that Total Internal Reflection takes place at $B$ is




A. $\sqrt{\frac{3}{2}}$

B. $\sqrt{3}$

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A. $\sqrt{\frac{3}{2}}$

B. $\sqrt{3}$

C. $\sqrt{\frac{4}{3}}$

D. None of these

**Question No. 15**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

A light beam exerts a force of $4.8 \times 10^{-4}$ N on a surface with an area of $0.04\ \mathrm{m}^2$, which absorbs $80\%$ of the incident light energy and remaining gets transmitted. If the speed of light is $3 \times 10^8$ m/s, calculate the light intensity in $W/\mathrm{cm}^2$.

A. $300\ \mathrm{W/cm^2}$

B. $400\ \mathrm{W/cm^2}$

C. $450\ \mathrm{W/cm^2}$

D. $500\ \mathrm{W/cm^2}$

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Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

There are four light-weight-rod samples A, B, C, D separately suspended by threads. A bar magnet is slowly brought near each sample and the following observations are noted

(i) A is feebly repelled

(ii) B is feebly attracted

(iii) C is strongly attracted

(iv) D remains unaffected

Which one of the following is true?

A. B is of a paramagnetic material

B. C is of a diamagnetic material

C. D is of ferromagnetic material

D. A is of a non-magnetic material

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A gas satisfies the relation $PV^{5/3} = K$ where $P$ is pressure, $V$ is volume and $K$ is constant. The dimensions of constant $K$ are

A. $\mathrm{ML^4T^{-2}}$

B. $\mathrm{ML^2T^{-2}}$

C. $\mathrm{ML^6T^{-2}}$

D. $\mathrm{MLT^{-2}}$

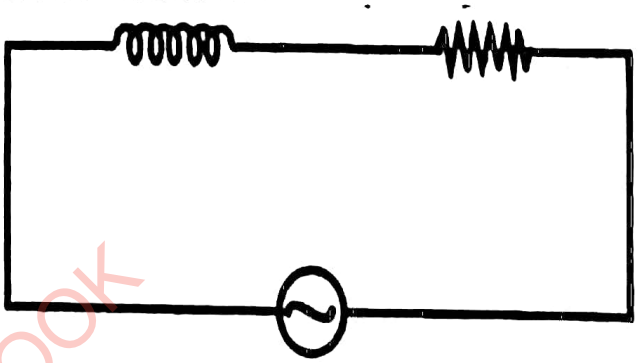
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Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In a series LR circuit connected to an alternating voltage source, it is observed that at the instant voltage across the source is maximum, voltage across the inductor is $3\ \mathrm{V}$ and voltage across the resistance is $4\ \mathrm{V}$. If resistance is $2\Omega$, what is reactance (in $\Omega$) of the inductor ?



A. $\sqrt{3}$

B. $\sqrt{2}$

C. $2\sqrt{2}$

D. $2\sqrt{3}$

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A wheel of moment of inertia 2.5 Kg-$\mathrm{m}^2$ has an initial angular velocity of 40 rads $^{-1}$. A constant torque of 10 Nm acts on the wheel. The time during which the wheel is accelerated to 60 $\mathrm{rads}^{-1}$ is

A. 6 s

B. 5 s

C. 4.5 s

D. 4 s

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Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Identify the correct statement(s) from among the following
(i) The constancy of the binding energy per nucleon in the range $30 < A < 170$ is a consequence of the fact that the nuclear force is short ranged.
(ii) The nuclear force does not depend on the charge of nucleons.
(iii) The nuclear force is repulsive when distance between two nucleons is less than $0.8\text{fm}$

A. (i) only

B. (ii), (iii)

C. (i), (ii)

D. (i), (ii), (iii)

## Physics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

The length of an open organ pipe is twice the length of another closed organ pipe. The fundamental frequency of the open pipe is $100\text{Hz}$. The frequency of the third harmonic of the closed pipe is

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

In gravity free space, a bead of charge $1\mu\text{C}$ and mass $3\text{mg}$ is threaded on a rough rod of friction coefficient $\mu = 0.3$. A magnetic field of magnitude $0.2\ \text{T}$ exists perpendicular to the rod. The bead is projected along the rod with a speed of $4\ \text{m/s}$. How much distance (in m ) will the bead cover before coming to rest?

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Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

A proton is fired from very far away towards a nucleus with charge $Q = 120\ e$, where, $e$ is the electronic charge. It makes the closest approach of 10 fm to the nucleus. The de Broglie wavelength (in units of fm) of the proton at its start is :
(take the proton mass, $m_p = \left(\frac{5}{3}\right) \times 10^{-27}\text{kg}$; $\frac{h}{e} = 4 \cdot 2 \times 10^{-15}$

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

A circular coil of area $0.01\ \text{m}^2$ and 40 turns is rotated about its vertical diameter with an angular speed of $50\ \text{rads}^{-1}$ in a uniform horizontal magnetic field $0.05\ \text{T}$. If the average power loss due to joule heating is $25\ \text{mW}$, then the closed loop resistance of the coil is

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

A block of mass $m = 0.3\ \text{kg}$ is connected to a spring of unknown spring constant $k$. It is compressed to a distance $x$ from its equilibrium position and released from rest. After approaching half the distance $\left(\frac{x}{2}\right)$ from equilibrium position, it hits another identical block and combines to move with a common velocity $2ms^{-1}$. The total initial energy of the spring is E. find 10E.

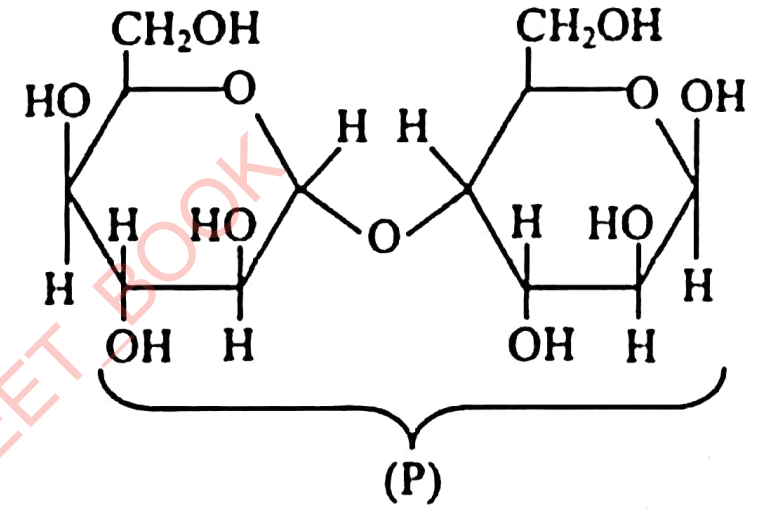
## Chemistry Single Correct (Maximum Marks: 80)

**Question No. 1**

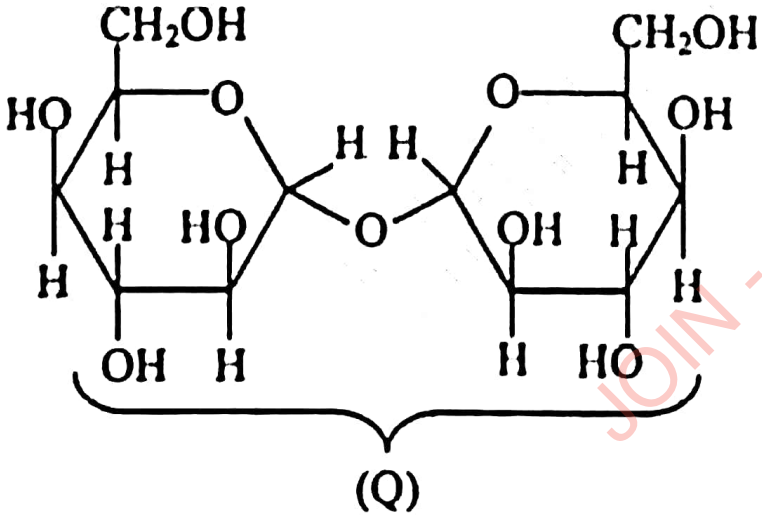
**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Which of the following disaccharide will not reduce Tollen's reagent?



(P)



(Q)

A. P

B. Q

C. P and Q both

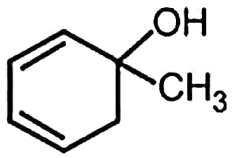
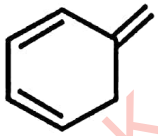
D. None of these

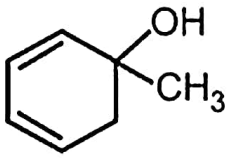
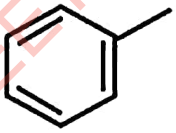
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Question No. 2

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Statement I :  undergo dehydration more easily to give  as a minor product and  as a major product.

Statement II :  is a non aromatic compound while  is an aromatic compound.

A. Both statement I and statement II are true and the statement II is the correct explanation of the statement I

B. Both statement I and statement II are true but the statement II is not the correct explanation of the statement I

C. Statement I is true but statement II is false

D. Both statement I and statement II are false

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Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the complex with the formula $MCl_3 \cdot 4H_2O$, the coordination number of the metal M is six and there is no molecule of hydration in it. The volume of $0.1$ M $AgNO_3$ solution needed to precipitate the free chloride ions in $200$ mL of $0.01$ M solution of the complex is

A. $40$ mL

B. $20$ mL

C. $60$ mL

D. $80$ mL

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For the cell $Zn\ (s)|ZnSO_4\ (aq)||CuSO_4\ (aq)|Cu\ (s)$, when the concentration of $Zn^{2+}$ is 10 times the concentration of $Cu^{2+}$, the expression for $\Delta G\ \left(\text{in J mol}^{-1}\right)$ is

[F =Faraday's constant, R =universal gas constant, T = temperature, $E^{\circ}cell = 1.1V$]

A. $2.303 \times RT + 1.1\ F$

B. $2.303 \times RT - 2.2\ F$

C. $1.1\ F$

D. $-2.2\ F$

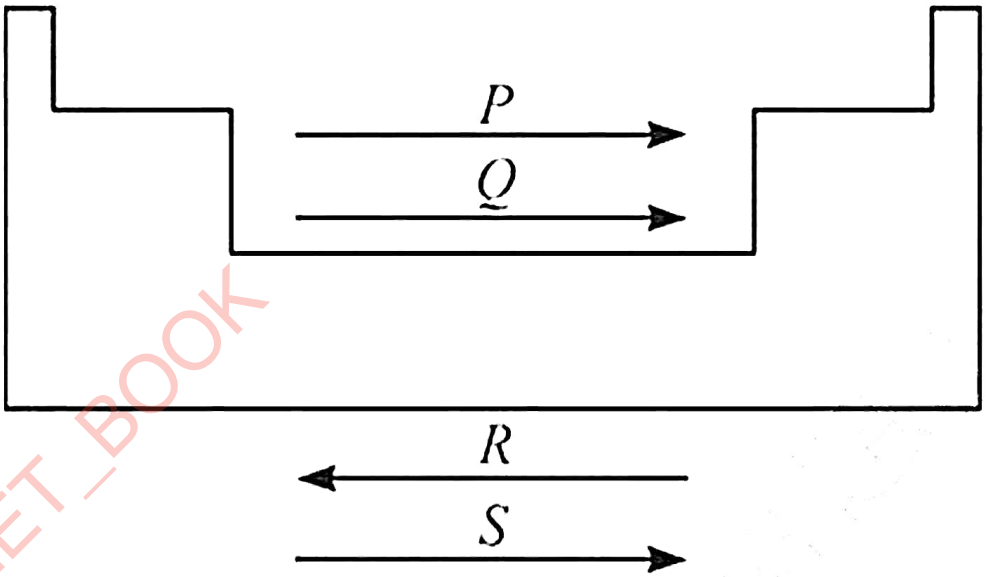
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Question No. 5

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Choose the correct combination for the given figure.




| | P | Q | R | S |
|---|---|---|---|---|
| A. | Ionisation enthalpy | Electron gain enthalpy | Atomic radius | Electro-negativity |
| B. | Atomic radius | Ionisation enthalpy | Electro-negativity | Electron gain |
| C. | Ionisation enthalpy | Atomic radius | Electro-negativity | Electron gain |
| D. | Electro-negativity | Electron gain enthalpy | Ionisation enthalpy | Atomic radius |

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Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following is not a disproportionation reaction?

A. $Hg_2Cl_2 \longrightarrow Hg + HgCl_2$

B. $4H_3PO_3 \longrightarrow 3H_3PO_4 + PH_3$

C. $Cl_2 + 2NaOH \longrightarrow NaCl + NaOCl + H_2O$

D. $BaO_2 + H_2SO_4 \longrightarrow BaSO_4 + H_2O_2$

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following statements are correct?

a. $Fe^{2+}$ ion can reduce $S_2O_8^{2-}$ to $SO_4^{2-}$.
b. KCl is oxidized by $Na_2Cr_2O_7$.
c. $\Delta G$ for $Mn^{3+}/Mn^{2+}$ is more negative than for $Cr^{3+}/Cr^{2+}$.
d. $VCl_5$ and $VBr_5$ are stable in the air.

A. Both b and c

B. Both b and d

C. Both a and c

D. a, b and c

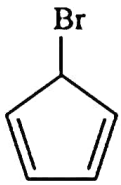
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Question No. 8

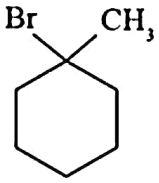
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

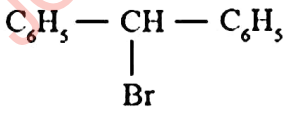
Consider the $S_N1$ solvolysis of the following halides in aqueous formic acid?



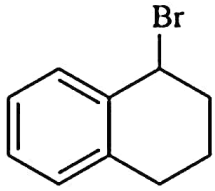

I.




II.




III.




IV.

Which one of the following is the correct sequence of the halides given above in the decreasing order of their reactivity?

A. III > IV > II > I

B. II > IV > I > III

C. I > II > III > IV

D. III > I > II > IV

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Match the molecules given from (List I) with the shape obtained through VSEPR theory (List II).

| | List (I) Molecules | | List (II) Shapes |
|---|---|---|---|
| A | $PCl_5$ | i | V-Shaped |
| B | $OF_2$ | ii | Triangular planar |
| C | $BCl_3$ | iii | Trigonal bipyramidal |
| D | $NH_3$ | iv | Trigonal pyramidal |
| | | v | Tetrahedral |

A. A – i, B – v, C – iv, D – iii

B. A – ii, B – iii, C – i, D – ii

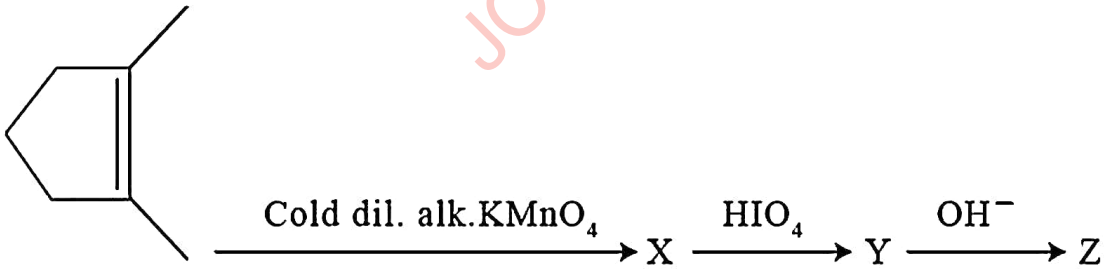
C. A – iv, B – iii, C – ii, D – v

D. A – iii, B – i, C – ii, D – iv

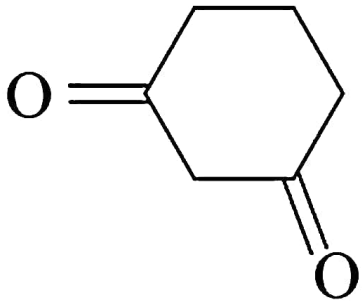
Question No. 10

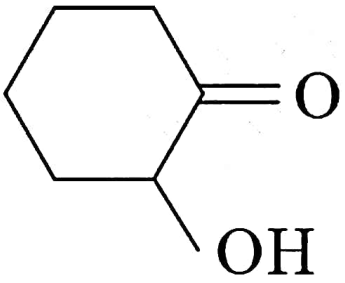
Only One Option Correct Type

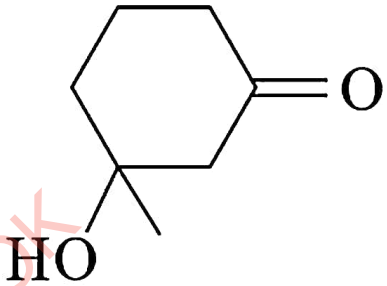
Each question has multiple options out of which ONLY ONE is correct.

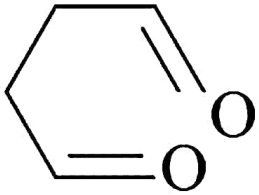



In the above sequence of reactions, $Z$ is

A. 

B. 

C. 

D. 

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Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The green coloured complex ion of nickel in its aqueous solution is

A. $\left[\mathrm{Ni(en)_3}\right]^{2+}$

B. $\left[\mathrm{Ni(H_2O)_2(en)_2}\right]^{2+}$

C. $\left[\mathrm{Ni(H_2O)_4(en)}\right]^{2+}$

D. $\left[\mathrm{Ni(H_2O)_6}\right]^{2+}$

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

How many litres of oxygen (at STP) are required for complete combustion of 39 g of liquid benzene?
(Atomic weights: $\mathrm{C} = 12, \mathrm{O} = 16, \mathrm{H} = 1$)

A. 84

B. 22.4

C. 42

D. 11.2

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Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Match the following lists:

| List –I | List –II |
|---|---|
| (A) Tartaric acid | (P) Metamerism |
| (B) $CH_3CH = CH - CH_3$ | (Q) Cis-trans isomerism |
| (C) $CH_3 - CH_2 - NO_2$ | (R) Optical isomerism |
| (D) $CH_3 - CH_2 - O - CH_2 - CH_3$ | (S) Conformational isomerism |
| | (T) Dynamic isomerism / Tautomerism |

A. $(A - R)(B - Q)\ (C - T)(D - P)$

B. $(A - P)(B - Q)\ (C - T)\ (D - S)$

C. $(A - R)(B - T)(C - Q)(D - P)$

D. $(A - R)(B - S)(C - T)(D - P)$

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion (A) $[SiF_6]^{2-}$ is formed but $[SiCl_6]^{2-}$ is not

Reason (R) Electronegativity (EN) of F is higher than EN of Cl

A. Both (A) and (R) are correct and (R) is the correct explanation of (A)

B. Both (A) and (R) are correct but (R) is not the correct explanation of (A)

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(A) is correct but (R) is
C. not correct

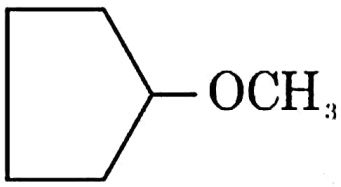
(A) is not correct but
D. (R) is correct

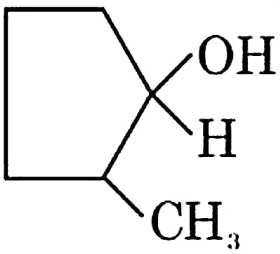
**Question No. 15**

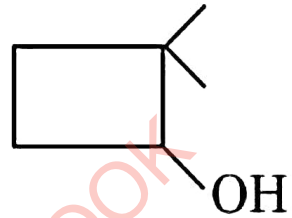
**Only One Option Correct Type**

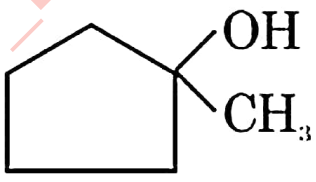
Each question has multiple options out of which ONLY ONE is correct.

An organic compound A (Molecular formula $C_6H_{12}O$ ) does not change the colour of acidic dichromate solution Compound A on treatment with $H_2SO_4$ produces alkene, which on oxidative ozonolysis gives a molecule $(C_6H_{10}O_3)$ which gives positive iodoform test. Find out the structure of ' A '.

A. 


B. 


C. 


D. 

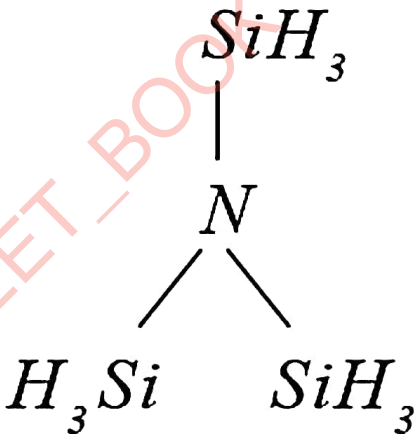

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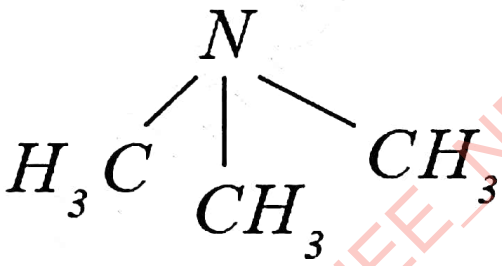
Question No. 16

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Trimethylammonia is pyramidal whereas trisilylammonia is a planar molecule. Why?




A. In silicon atom, the electron pair of nitrogen is involved in $p\pi - d\pi$ bonding.

B. In nitrogen, lone pair of electrons is present at nitrogen hence it has pyramidal shape.

C. Both (a) and (b) are correct.

D. Both (a) and (b) are wrong.

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Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

1 g of a complex $\left[Cr(H_2O)_5Cl\right]Cl_2 \cdot H_2O$ (mol. Wt. 266.5) was passed through a cation exchanger to produce HCl. The acid liberated was diluted to 1 litre. The normality of this acid solution is

A. $5 \times 10^{-3}$ N

B. $7.5 \times 10^{-3}$ N

C. $7.5 \times 10^{-2}$ N

D. $7.5 \times 10^{-1}$ N

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the borax bead test of $Co^{2+}$, the blue colour of bead is due to the formation of:

A. $B_2O_3$.

B. $Co_3B_2$.

C. $Co(BO_2)_2$.

D. $CoO$.

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Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The equilibrium constant for the reaction

$SO_2(g) + \frac{1}{2} O_2(g) \rightleftharpoons SO_3(g)$ is $5 \times 10^{-2}$ atm.

The equilibrium constant of the reaction

$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$ would be

A. 100 atm

B. 200 atm

C. $4 \times 10^2$ atm

D. $6.25 \times 10^4$ atm

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Observe the following complexes

$[Mn(CN)_6]^{3-}$, $[Fe(CN)_6]^{-3}$, $[Co(C_2O_4)_3]^{3-}$, $[MnCl_6]^{3-}$,

$[Fe(CN)_6]^{4-}$, $[CoF_6]^{3-}$

From the above, the number of inner orbital complexes with paramagnetic nature is

A. 2

B. 4

C. 1

D. 3

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## Chemistry Numerical (Maximum Marks: 20)

**Question No. 1**

**Numerical Type**

The answer has to be filled into the input box provided below.

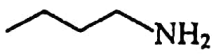
1.0 g of monobasic acid $HA$ in 100 g water lowers the freezing point by 0.145 K. If 0.45 g of same acid on 1000 mg of water requires 27 mL of $\frac{N}{9}$ NaOH solution for complete neutralisation, then % degree of dissociation of acid is ..... $\%\left(K_f(H_2O) = 1.86 \text{ K kg mol}^{-1}\right)$ (mark answer to nearest whole number)

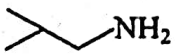
**Question No. 2**

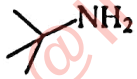
**Numerical Type**

The answer has to be filled into the input box provided below.

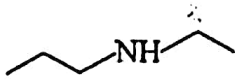
x = Number of compound give $3^\circ$ alcohol when reacts with $HNO_2$ ?

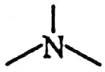
(a) 

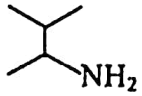
(b) 

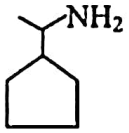
(c) 

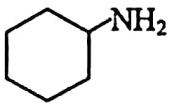
(d) $Ph - CH_2 - NH_2$

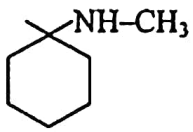
(e) 

(f) 

(g) 

(h) 

(i) 

(j) 

**Question No. 3**

**Numerical Type**

The answer has to be filled into the input box provided below.

In the reaction of NO by $H_2$ it was observed that equimolecular mixture of gases at 500 mm was half changed in 100 seconds. In another experiment with an initial pressure of 250 mm, the reactions was half completed in 200 seconds. Calculate the order of reaction-

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

In neutral or faintly alkaline solution, 8 moles of permanganate anion quantitatively oxidize thiosulphate anions to produce X moles of a sulphur containing product. The magnitude of X is

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

The number of electron that can be present in sub-shells having $m_s$ value of $\frac{-1}{2}$, for $n$ up to 3.

## Mathematics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{b} = -3\hat{i} + \hat{j} - \hat{k}$ and $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$, $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$ then a unit vector in the direction of $\vec{r}$ is

A. $\frac{1}{3}(-2\hat{i} + \hat{j} - \hat{k})$

B. $\frac{1}{3}(-2\hat{i} - \hat{j} + 2\hat{k})$

C. $\frac{1}{3}(-2\hat{i} - \hat{j} - 2\hat{k})$

D. None of these

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Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

$A, B, C$ are vertices of a triangle with right angle at $A$ and $P(-4, 0); Q(0, 6)$ are two given points. If the ratio of distances from each vertex to $P$, to that of $Q$ is $2:3$, then the centroid of $\triangle ABC$ lies on a circle with radius equal to

A. $\frac{4\sqrt{13}}{5}$ units

B. 4 units

C. $\frac{8\sqrt{13}}{5}$ units

D. 8 units

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The sum of the roots of the equation $2^{33x-2} + 2^{11x+2} = 2^{22x+1} + 1$ is

A. $\frac{1}{11}$

B. $\frac{2}{11}$

C. $\frac{3}{11}$

D. $\frac{4}{11}$

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Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider two lines in space as $L_1 : \vec{r}_1 = \hat{j} + 2\hat{k} + \lambda\left(3\hat{i} - \hat{j} - \hat{k}\right)$ and $L_2 : \vec{r}_2 = 4\hat{i} + 3\hat{j} + 6\hat{k} + \mu\left(\hat{i} + 2\hat{k}\right)$ where $\lambda,\ \mu \in R$. If the shortest distance between these lines is $\sqrt{d}$. Then the value of $d$ will be:

A. $5$

B. $6$

C. $7$

D. $8$

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

$\lim_{x\to 0} \frac{\sin x^4 - x^4 \cos x^4 + x^{20}}{x^4\left(e^{2x^4} - 1 - 2x^4\right)}$ is equal to

A. $0$

B. $-\frac{1}{6}$

C. $\frac{1}{6}$

D. does not exist

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Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The indefinite integral $I = \int \frac{\left(\sin^2 x - \cos^2 x\right)^{2019}}{(\sin x)^{2021}(\cos x)^{2021}}\, dx$ simplifies to (where $c$ is an integration constant)

A. $\frac{\left(\sin^2 x - \cos^2 x\right)^{2020}}{2020} + c$

B. $\frac{(\tan x - \cot x)^{2020}}{2020} + c$

C. $\frac{(\sin x - \cos x)^{2020}}{2020} + c$

D. $\frac{\left(\tan^2 x + \cot^2 x\right)^{2020}}{2020} + c$

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The value of expression $\frac{\sin^4\alpha + \cos^4\alpha + 2\sin^2\alpha\cos^2\alpha}{\sin^6\alpha + \cos^6\alpha + 3\sin^2\alpha\cos^2\alpha} + 1$ is equal to:

A. 1

B. 3

C. 2

D. 0

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Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Three ordinary and fair dice are rolled simultaneously. The probability of the sum of outcomes being atleast equal to 8, is equal to

A. $\frac{81}{108}$

B. $\frac{27}{216}$

C. $\frac{81}{216}$

D. $\frac{181}{216}$

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The general solution of the differential equation $\left(x^2+2\right)$ $dy+2xydx=e^x\left(x^2+2\right)dx$ is

A. $\frac{x}{y}=e^x\left(x^2+x-4\right)+c$

B. $2xy=e^x\left(x^2-2x+4\right)+c$

C. $\left(x^2+2\right)y=e^x\left(x^2-2x+4\right)+c$

D. $\left(x^2+2\right)^2y=e^x\left(x^2+2x-4\right)+c$

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Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider the sequence of numbers $\alpha_0, \alpha_1, \ldots, \alpha_n$ where $\alpha_0 = 17.23, \alpha_1 = 33.23$ and $\alpha_{r+2} = \frac{\alpha_r + \alpha_{r+1}}{2}$. Then select incorrect option

A. $|\alpha_{10} - \alpha_9| = \frac{1}{32}$

B. $\alpha_0 - \alpha_1, \alpha_1 - \alpha_2, \alpha_2 - \alpha_3, \ldots$ are in G.P.

C. $\alpha_0 - \alpha_2, 2(\alpha_1 - \alpha_2), \alpha_1 - \alpha_3$ are in H.P.

D. $|\alpha_{10} - \alpha_9| = |\alpha_8 - \alpha_7|$

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $A = \{1, 2, 3, 4, 5, 6\}$ number of functions $f$ from $A$ to $A$ such that $f(m) + f(n) = 7$, whenever $m + n = 7$ is

A. 288

B. 216

C. 144

D. 196

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Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ and a non-singular matrix $P = \begin{bmatrix} -1 & 1 \\ a & b \end{bmatrix}$ is such that $P^{-1}AP = \text{diag}(-1, 3)$, then the sum of all the elements of $A^{100}$ is

A. $4 \times 3^{100}$

B. $2 \times 3^{100}$

C. $3^{100}$

D. $0$

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $f(x) = \begin{cases} x^3 + x^2 - 10x & , \quad -2 \leq x < 0 \\ \sin x & , \quad 0 \leq x < \frac{\pi}{2} \\ 1 + \cos x & , \quad \frac{\pi}{2} \leq x < \pi \end{cases}$, then $f(x)$

A. has a local maxima at $x = 0$ and $x = \frac{\pi}{2}$

B. has a local minima at $x = \frac{\pi}{2}$

C. has a local minima at $x = \frac{\pi}{2}$ and a local maxima at $x = 0$

D. has a local minima at $x = 0$ and a local maxima at $x = \frac{\pi}{2}$

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Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In a right angled triangle $ABC, BC = 5, AB = 4, AC = 3$, Let $S$ be the circum circle of triangle ABC and $S_1$ be the circle touching both sides AB and AC and circle S internally. Let $S_2$ be the circle touching the sides AB and AC of $\Delta^{le}$ ABC, and touching the circle $S$ externally if $r_1, r_2$ are radii of circles $S_1$ and $S_2$ respectively, then $\sqrt{1 + r_1 r_2}$ is

A. 3

B. 6

C. 4

D. 5

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $H : \frac{(5x-y-24)^2}{26} - (x + 5y - 10)^2 = 26$ is a hyperbola, then

A. length of transverse axis is $\sqrt{26}$.

B. length of conjugate axis is 1.

C. center is $(1, 5)$

D. one vertex is $(0, 2)$

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

$A = \begin{bmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{bmatrix}$ and $B = \begin{bmatrix} p_1 & q_1 & r_1 \\ p_2 & q_2 & r_2 \\ p_3 & q_3 & r_3 \end{bmatrix}$ $p_i$ , $q_i$ , $r_i$ are the cofactors of the elements $l_i$, $m_i$, $n_i$ for $i = 1$ ,2, 3. If $(l_1, m_1, n_1), (l_2, m_2, n_2)$ and $(l_3, m_3, n_3)$ are the direction cosines of three mutually perpendicular lines, then $(p_1, q_1, r_1), (p_2, q_2, r_2)$ and $(p_3, q_3, r_3)$ are

A. the direction cosines of three mutually perpendicular lines

B. the direction ratios of three mutually perpendicular lines which are not direction cosines

C. the direction cosines of three lines which need not be perpendicular

D. the direction ratios but not the direction cosines of three lines which need not be perpendicular

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $f(x) = \sin 2x - 8(a+1)\sin x + (4a^2 + 8a - 14)x$ increases for all $x \in R$ and has no critical points for all $x \in R$, then which of the following values of $a$ are not possible?

A. $(-\infty, -\sqrt{5} - 2)$

B. $(1, \infty)$

C. $(\sqrt{5}, \infty)$

D. None of these

**Question No. 18**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Let z and w be non-zero complex numbers such that $zw = |z^2|$ and $|z - \overline{z}| + |w + \overline{w}| = 4$. If w varies, then the perimeter of the locus of z is

A. $8\sqrt{2}$ units

B. $4\sqrt{2}$ units

C. 8 units

D. 4 units

**Question No. 19**

**Only One Option Correct Type**

Each question has multiple options out of which ONLY ONE is correct.

Consider $I_1 = \int_0^{\pi/2} (\cos x)^{\sqrt{17}+3} dx$ and $I_2 = \int_0^{\pi/2} (\cos x)^{\sqrt{17}+5} dx$, then $\frac{I_1}{I_2}$ is equal to-

A. $\frac{\sqrt{17}+3}{\sqrt{17}+5}$

B. $\frac{8}{\sqrt{17}+3}$

C. $\sqrt{17} + 3$

D. None of these

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Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Find the area bounded by the curves,
$x^2 + y^2 = 25, 4y = |4 - x^2|$ and $x = 0$ in the first quadrant.

A. $1 + \sin^{-1}\left(\frac{4}{5}\right)$

B. $2 + 25\sin^{-1}\left(\frac{4}{5}\right)$

C. $2 + \frac{25}{2}\sin^{-1}\left(\frac{4}{5}\right)$

D. None of these

## Mathematics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

If $206^{1753}$ is divided by 17, then find the remainder.

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

If

$$A = \begin{bmatrix} 1 & 25 \end{bmatrix}, B = \begin{bmatrix} 0 & \frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}, C = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}, D = \begin{bmatrix} 0 & \frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix},$$

and $F = [40]$ are matrices and $\alpha, \beta$ are roots of equation $AB^5C^{10}D^5E = F$ then the value of $(1-\alpha)(1-\beta)$ is

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

If

$$A = [1 \quad 25], B = \begin{bmatrix} 0 & \frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}, C = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}, D = \begin{bmatrix} 0 & \frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix},$$

and $F = [40]$ are matrices and $\alpha, \beta$ are roots of equation $AB^5C^{10}D^5E = F$ then the value of $(1-\alpha)(1-\beta)$ is

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

Let $F(x)$ be a real-valued function defined for all real $x \neq 0, 1$ such that

$$F(x) + F\left(\frac{x-1}{x}\right) = 1 + x.$$

Find value of $8F(2)$.

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

$a, b, c, d, e, f, g, h$ be distinct elements in the set $\{-7, -5, -3, -2, 2, 4, 6, 13\}$. The minimum value of $(a+b+c+d)^2 + (e+f+g+h)^2$ is

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

If $6p^2$ numbers of 4 digits can be made with the digit 0, 1, 2, 3, 4, 5 which are divisible by 3, digit being unrepeated in the same number, then find the value of p + 4.

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ied below.

$$\left|, \mathrm{D}=\begin{bmatrix} 0 & \frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}, \mathrm{E}=\begin{bmatrix} x^2-5x+20 \\ x+2 \end{bmatrix}\right.$$

equation
$-\beta)$ is

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